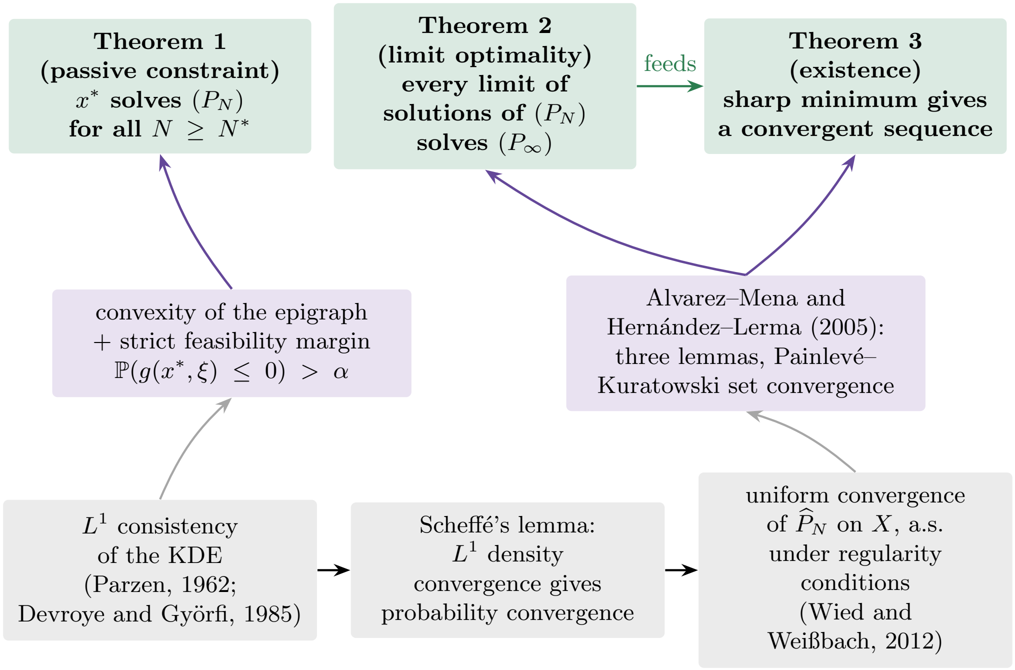

Theorem 1 (passive constraint) x^* **solves** (P_N) **for all** $N \geq N^*$
Theorem 2 (limit optimality) every limit of solutions of (P_N) **solves** (P_∞)
feeds
Theorem 3 (existence) sharp minimum gives a convergent sequence
convexity of the epigraph + strict feasibility margin $\mathbb{P}(g(x^*, \xi) \leq 0) > \alpha$
Alvarez–Mena and Hernández–Lerma (2005): three lemmas, Painlevé–Kuratowski set convergence
L^1 consistency of the KDE (Parzen, 1962; Devroye and Györfi, 1985)
Scheffé's lemma: L^1 density convergence gives probability convergence
uniform convergence of $\widehat{P}_N$ on X, a.s. under regularity conditions (Wied and Weißbach, 2012)